Formal verification of infinite state systems with unbounded agents

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05/03/2026

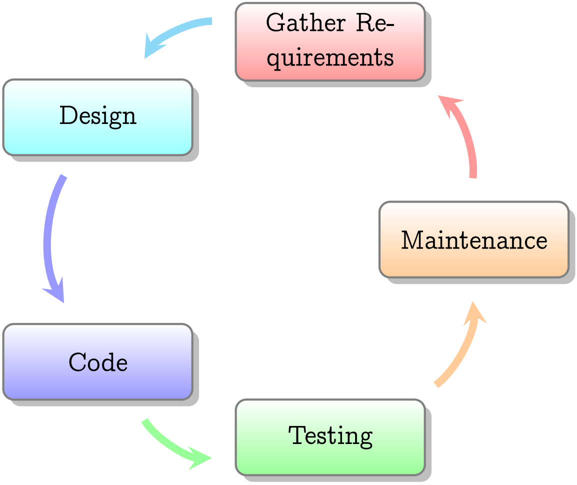
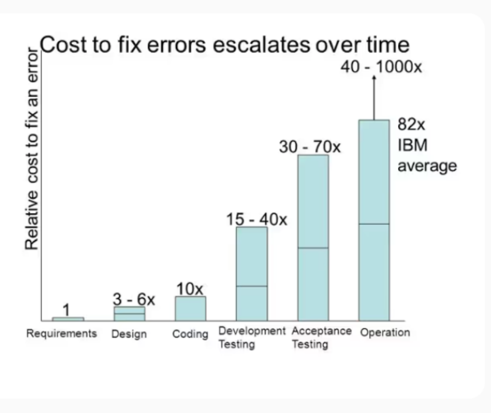


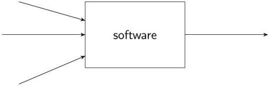
Figure: Traditional Software Development Life Cycle

Error Costs

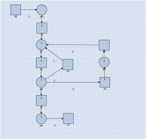
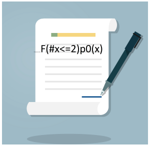


J. Stecklein, "Error cost escalation through the project life cycle, 2010

What is model checking?

Does 

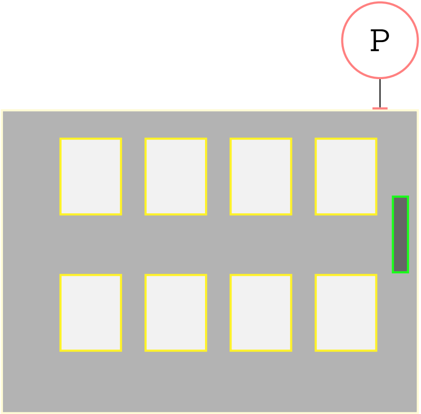
satisfy  ?

Does  satisfy 

?

Running example: Unbounded Client-Server System (UCS)

An empty Parking Lot waits for vehicles



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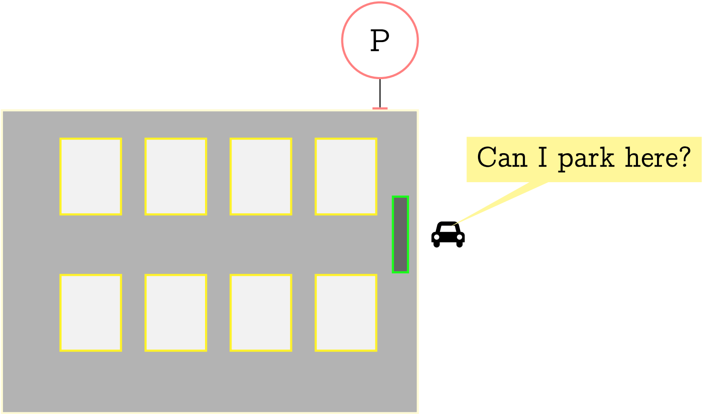

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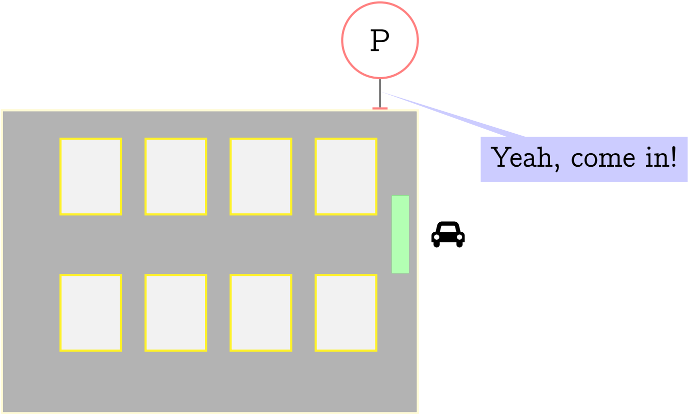

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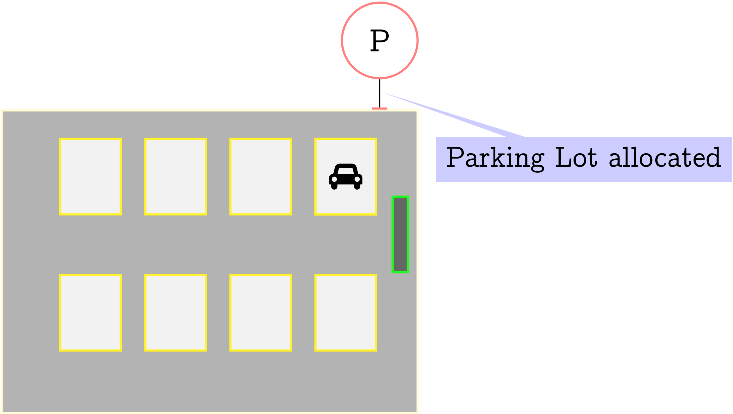
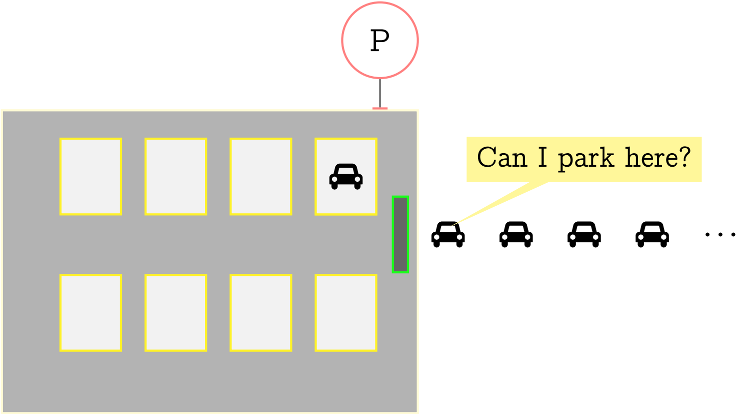


Running example: Unbounded Client-Server System (UCS)

Unbounded number of parking requests



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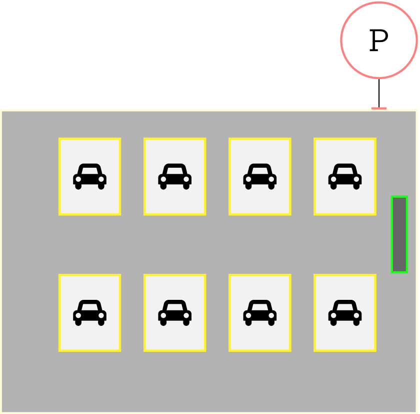
No parking space



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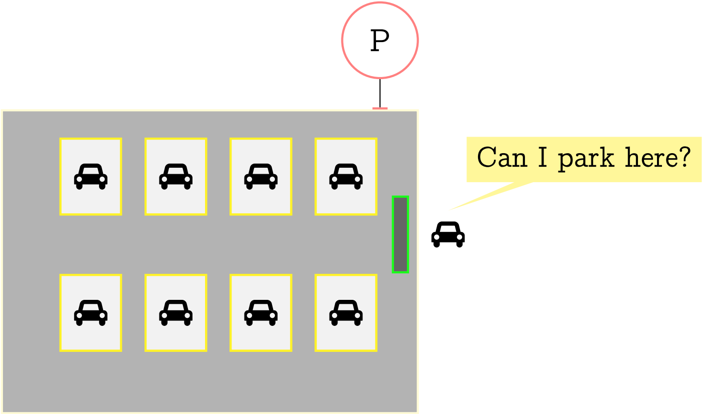

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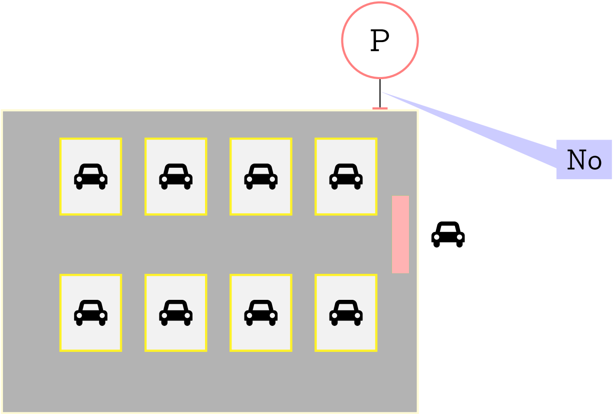

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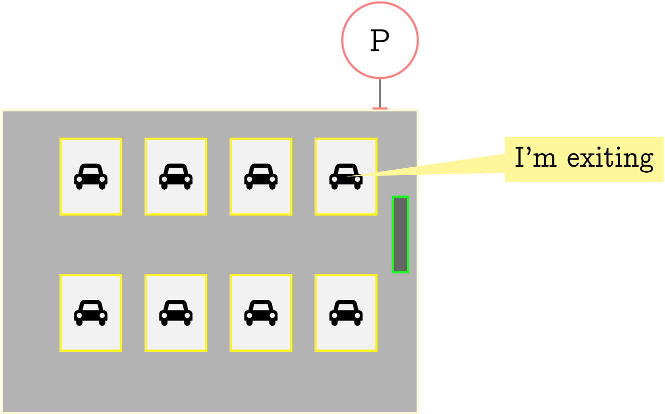

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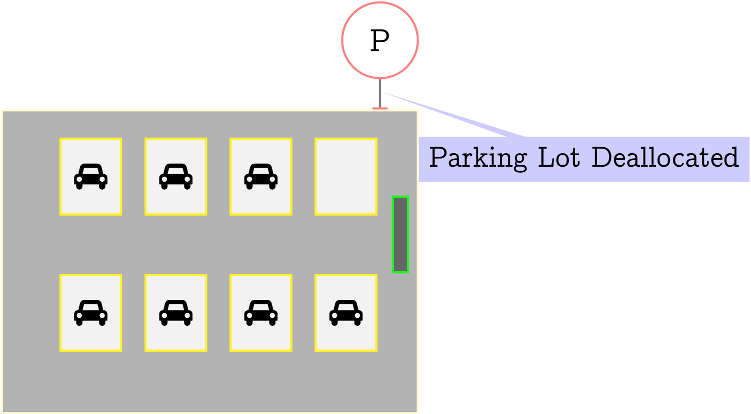

Running example: Unbounded Client-Server System (UCS)

- single server-multiple client system
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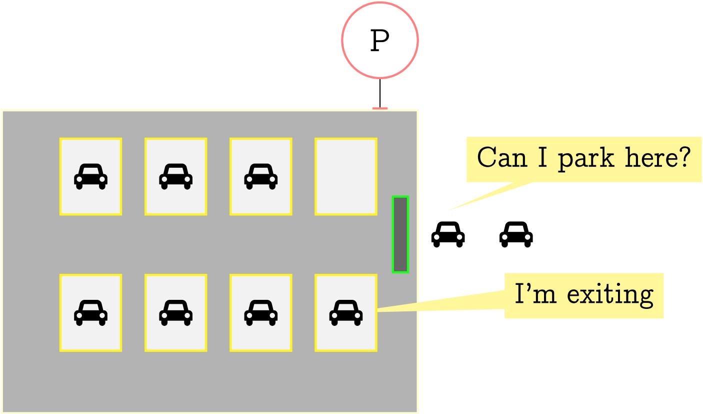
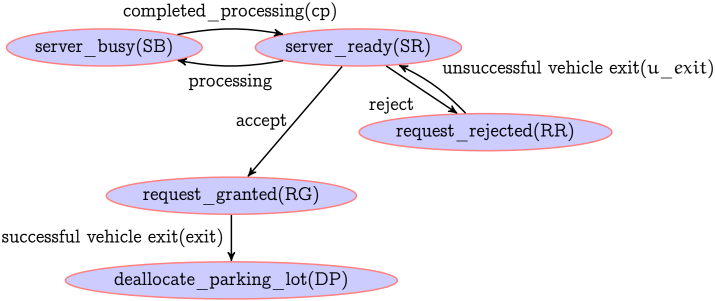
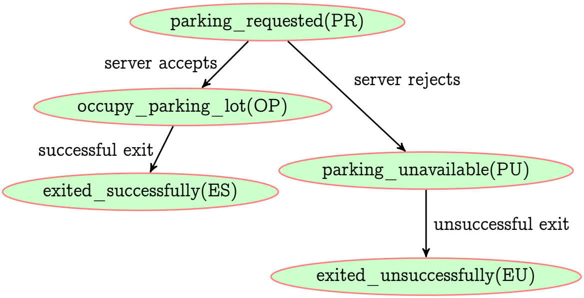


Case Study Parking System: Server behaviour



Case Study Parking System: Client behaviour



Research Questions

1. How do we formally verify UCS?
2. How do we formally model UCS?
3. How do we express properties of client-server systems naturally?
    1. Do existing logics suffice?
    2. What additional properties are interesting in the context of unbounded client-server systems?

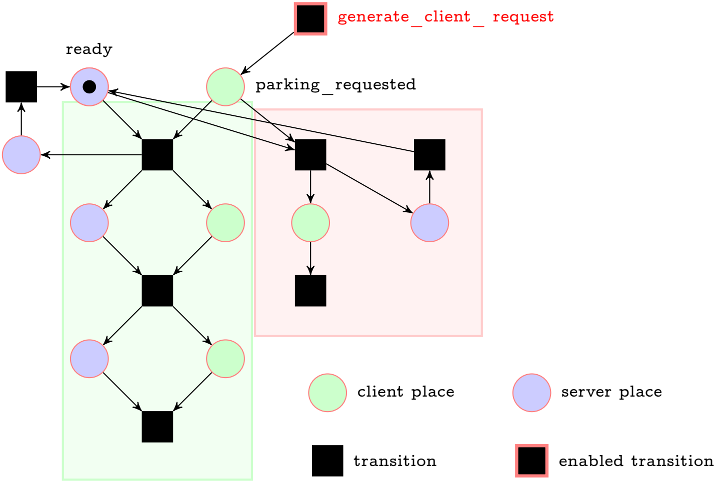
Research Questions

1. How do we formally verify UCS?
2. How do we formally model UCS?
3. How do we express properties of client-server systems naturally?
    1. Do existing logics suffice?
    2. What additional properties are interesting in the context of unbounded client-server systems?
4. Are there formal verification tools for UCS? Can we add to the repertoire?
    1. How are existing tools different, unique? Is another tool necessary?

Challenge 1: Modelling UCS using Petri Nets

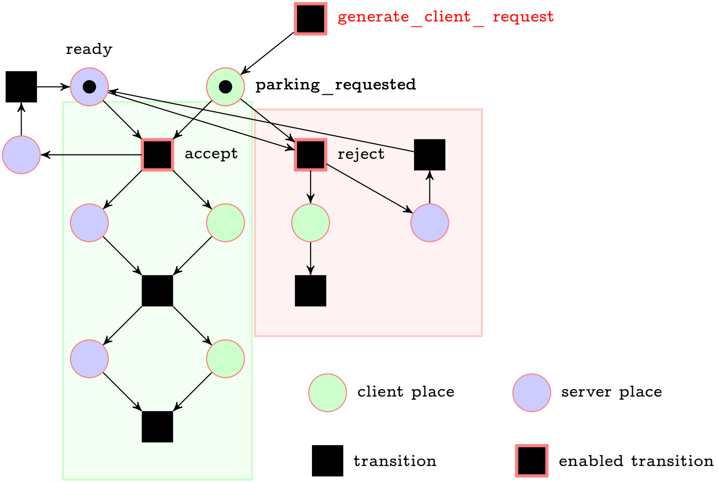
Initially the server is

ready for client requests



Challenge 1: Modelling UCS using Petri Nets

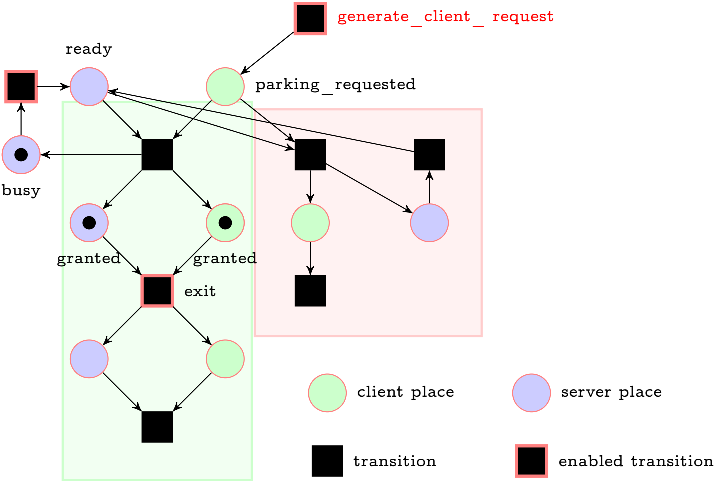
Client parking request is received



Challenge 1: Modelling UCS using Petri Nets

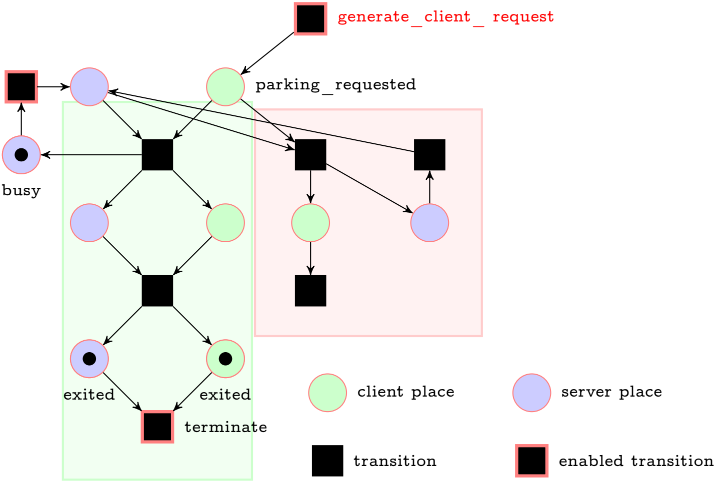
Server accepts the Client

parking request



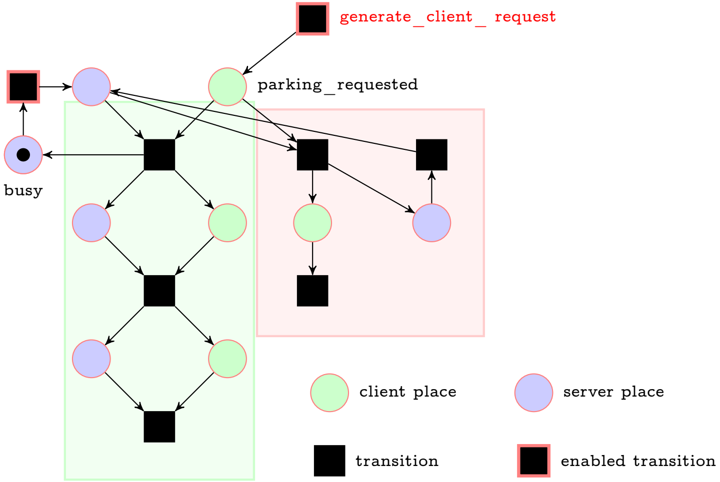
Challenge 1: Modelling UCS using Petri Nets

Client exits the parking space



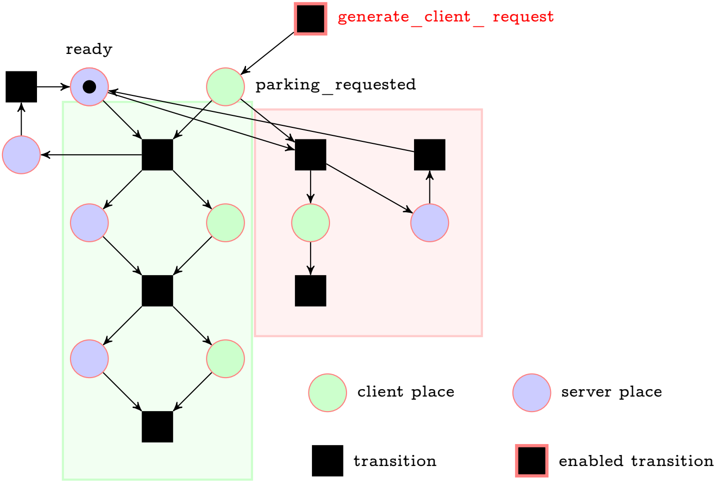
Challenge 1: Modelling UCS using Petri Nets

Client is terminated



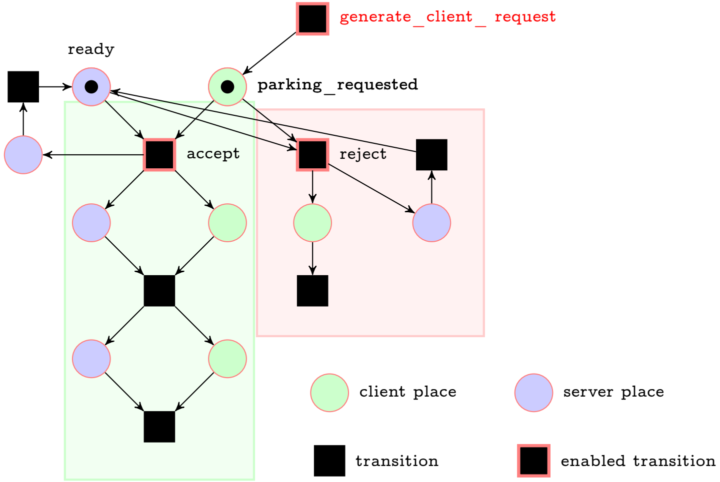
Challenge 1: Modelling UCS using Petri Nets

Server is ready for more requests



Challenge 1: Modelling UCS using Petri Nets

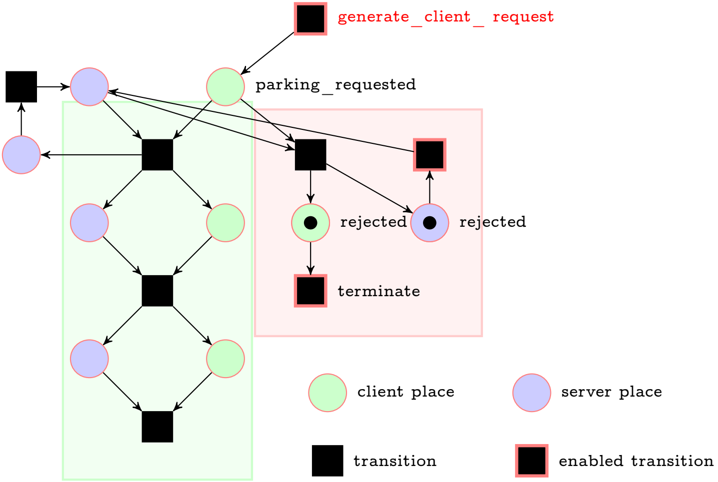
A new Client parking request is received



Challenge 1: Modelling UCS using Petri Nets

Server rejects the Client

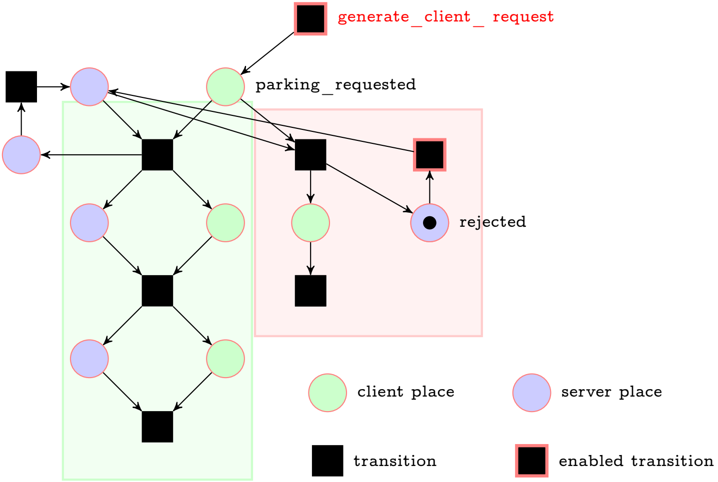
parking request



Challenge 1: Modelling UCS using Petri Nets

Client is terminated

unsuccessfully



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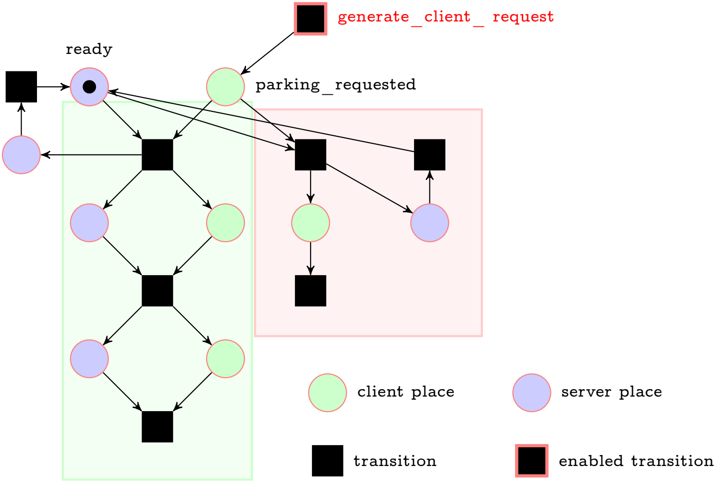

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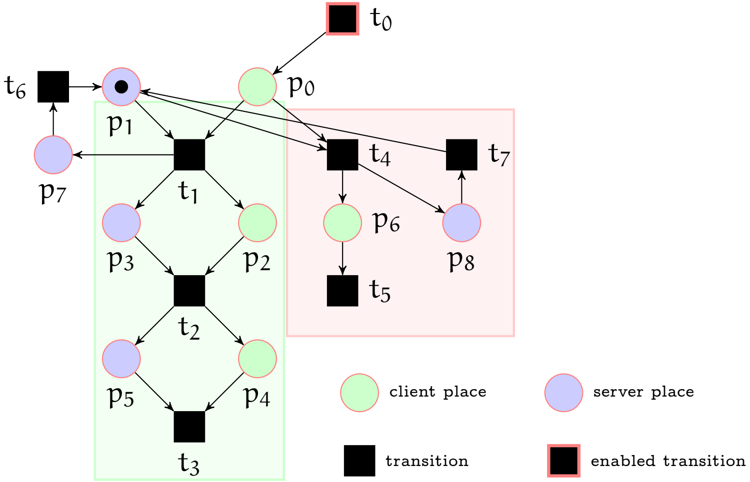

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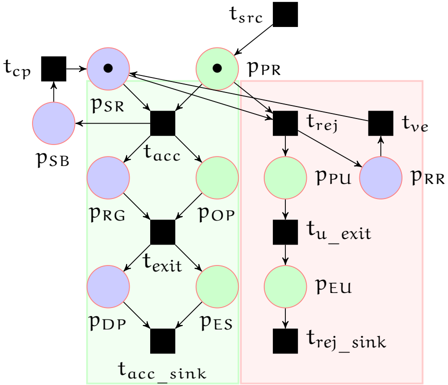

Finite sets of atomic client propositions P_c and server propositions P_s:

$P_c = \{$parking_requested(PR),
occupy_parking_lot(OP),
parking_unavailable(PU),
exit_successfully(ES),
exited_unsuccessfully(EU)$\}$

$P_s = \{$server_ready(SR),
server_busy(SB),
request_granted(RG),
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Properties

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   $G(\#p_{OP} = \#p_{RG})$

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3. Is the number of rejected requests greater than the number of accepted requests at some point?

   $F(\#p_{EU} > \#p_{ES})$.

Linear Temporal Logic with Integer Arithmetic LTL_{LIA}

In LTL_{LIA},there are two types of atomic formulas:

1. P_s - describing basic server properties, which are propositional constants and can be used to describe the transitions of the net
2. $\beta \in \Delta$, the set of client formulas

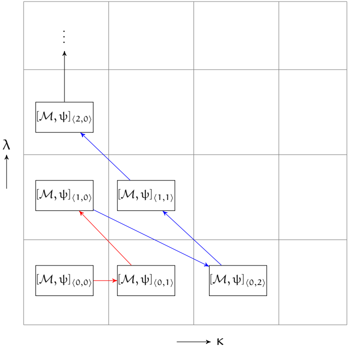
$$\alpha, \hat{\alpha} ::= \#p \mid \mathfrak{c} \mid (\mathfrak{c} * \#p) \mid (\#p * \mathfrak{c}) \mid (\alpha + \hat{\alpha}) \mid (\alpha - \hat{\alpha})$$
$$\beta ::= (\alpha < \hat{\alpha}) \mid (\alpha > \hat{\alpha}) \mid (\alpha \leq \hat{\alpha}) \mid (\alpha \geq \hat{\alpha}) \mid (\alpha = \hat{\alpha})$$

where $p \in P_c$, the set of client propositions and $\mathfrak{c} \in \mathbb{Z}^+$.
The set of server formulas Ψ are defined as follows:

$$\psi ::= q \in P_s \mid \beta \in \Delta \mid \neg\psi \mid \psi \vee \psi' \mid \psi \wedge \psi' \mid X\psi \mid F\psi \mid G\psi \mid \psi U \psi'$$

where $\psi, \psi' \in \Psi$. Modalities X, F, G and U are the usual modal operators: Next, Eventually, Globally and Until, respectively.

BMC vs 2D-BMC



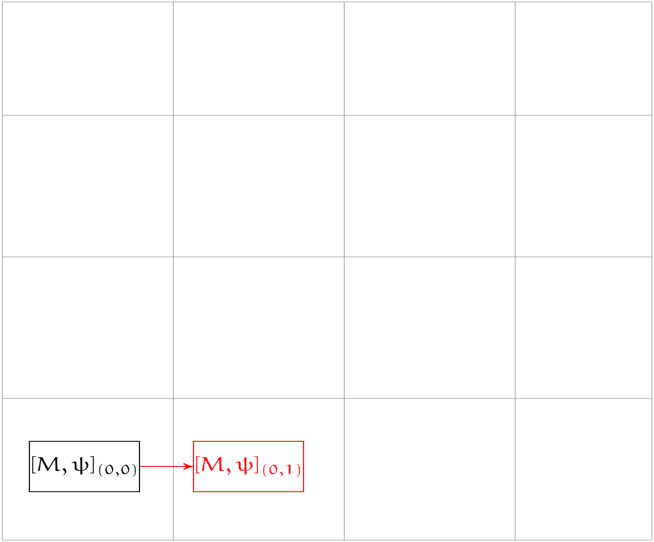
- In BMC, "We ask whether the system M has any counterexample of length k to (property) ψ. This bounded problem is encoded into SAT." - A.Biere[a]
- In 2D-BMC, we ask whether the system M has any counterexample to ψ, of length λ and number of tokens κ and encode this bounded problem into SMT.

[a] Linear Encodings of Bounded LTL Model Checking, Biere et al, LMCS, 2006

The 2D-BMC strategy

execution steps + number of clients = bound

The 2D-BMC strategy

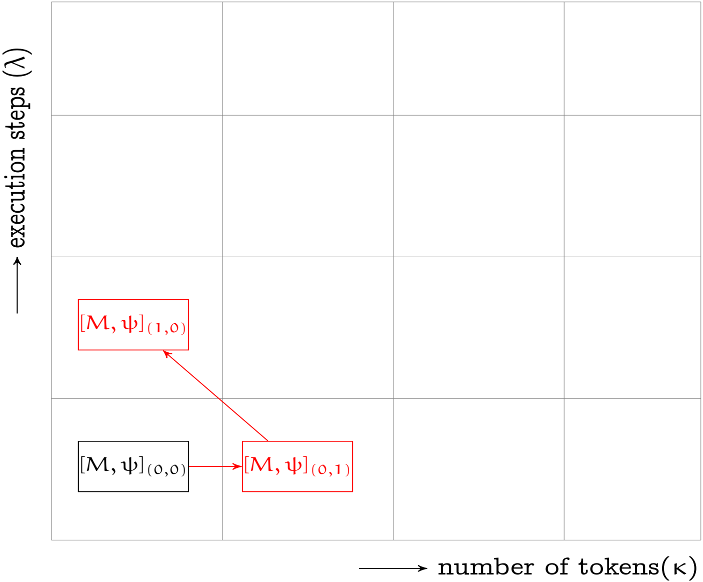


$\longrightarrow$ number of tokens(κ)

$\longrightarrow$ execution steps (λ)

execution steps + number of clients = bound

The 2D-BMC strategy



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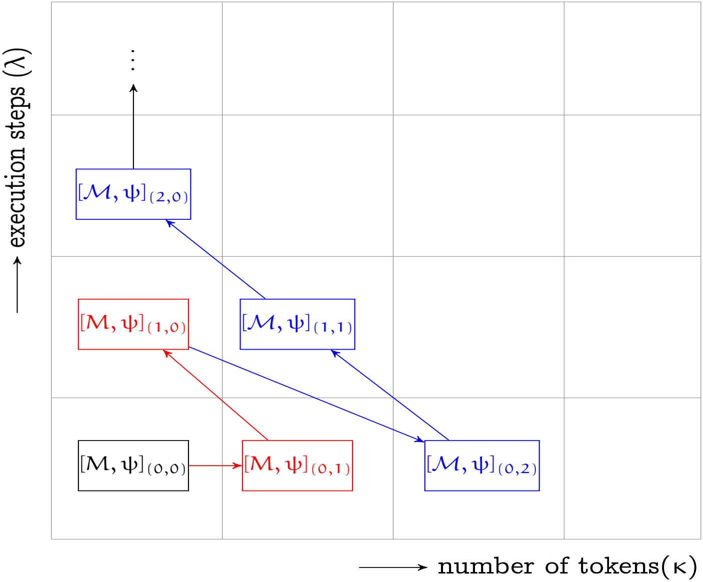

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Tool Architecture

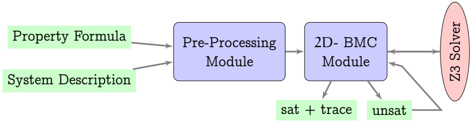


Figure: DCModelChecker architecture

Experimental Results

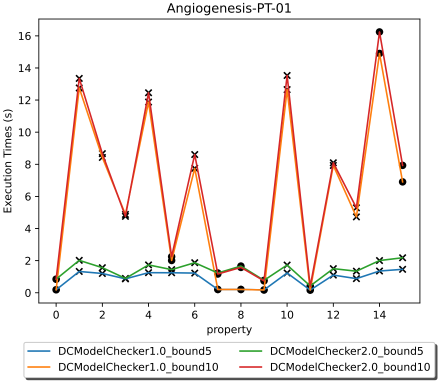


Figure: Verification of FireabilityCardinality Properties with bound 5 and 10

Verified against 16 Benchmarks from Model Checking Contest 2025

Experimental Results

Model	Execution Times(ms)			
	DCModelChecker 1.0	DCModelChecker 2.0	ITS-Tools	Tapaal
Parity	**0.010**	0.40	0.933	0.02
PGCD	**0.019**	0.98	1.487	0.07
Process	0.033	0.85	1.201	**1e-05**
CryptoMiner	0.036	0.64	0.957	**7e-06**
Murphy	0.031	0.83	1.161	**8e-06**

Table: Results of Comparative Experiments on Unbounded PNs

Contributions

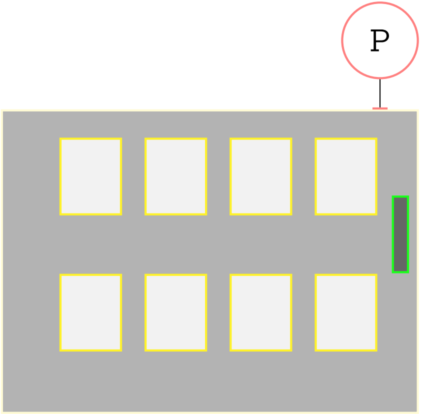
- Part 1: Verification of Unbounded Client-Server (UCS) using Petri Nets and Linear Temporal Logic with integer arithmetic using 2D-BMC

Challenge 2: Modelling an UCS with distinguishable clients

What if clients needed to be distinguished?

Running example

An empty Parking Lot waits for vehicles



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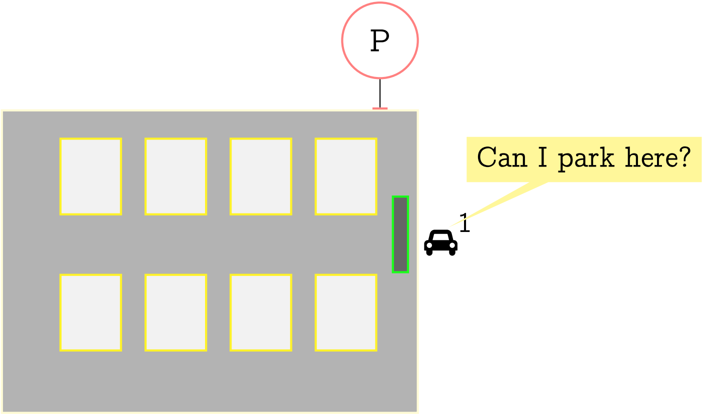

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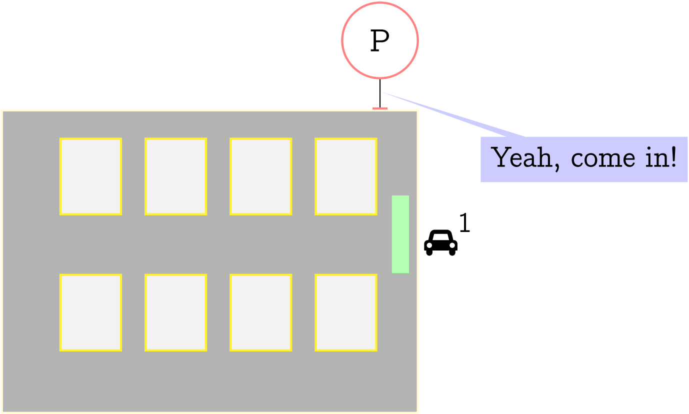

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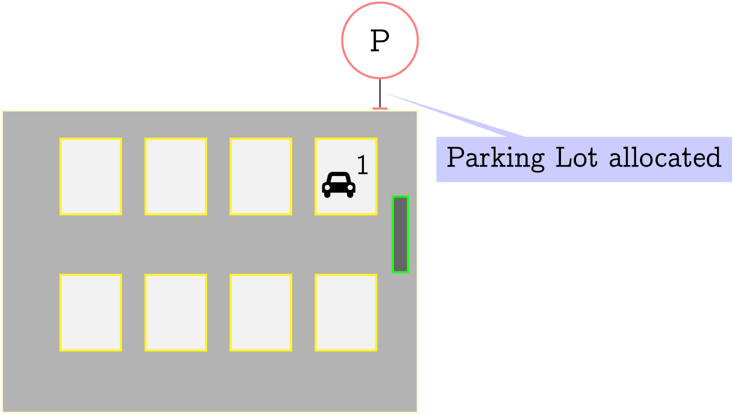
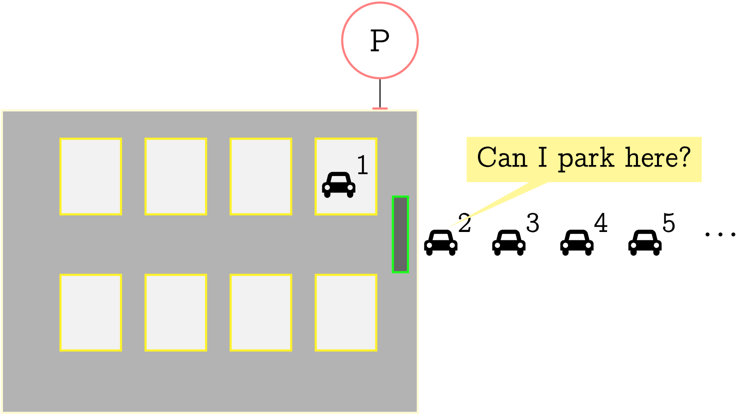


Running example

Unbounded number of parking requests



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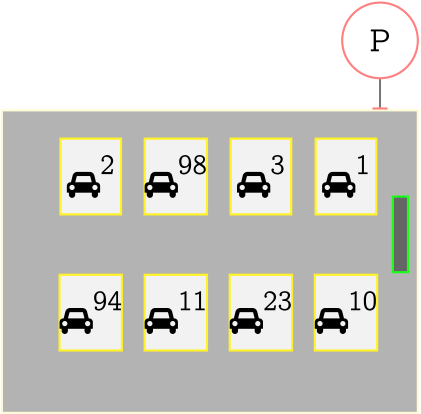

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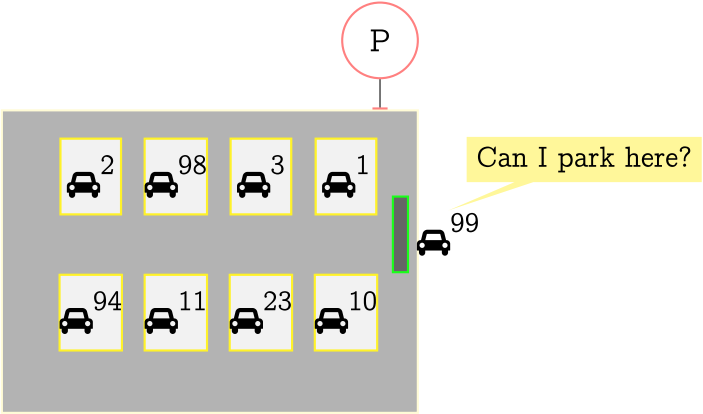

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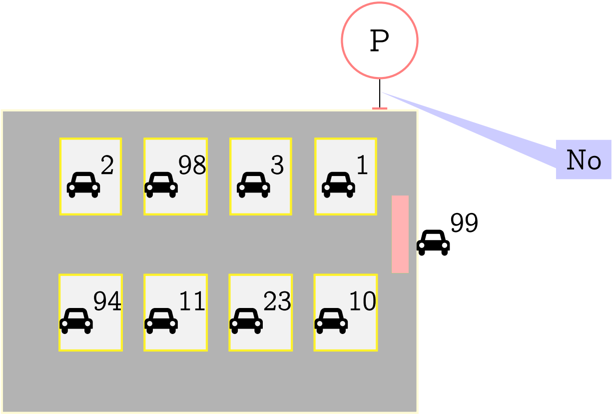

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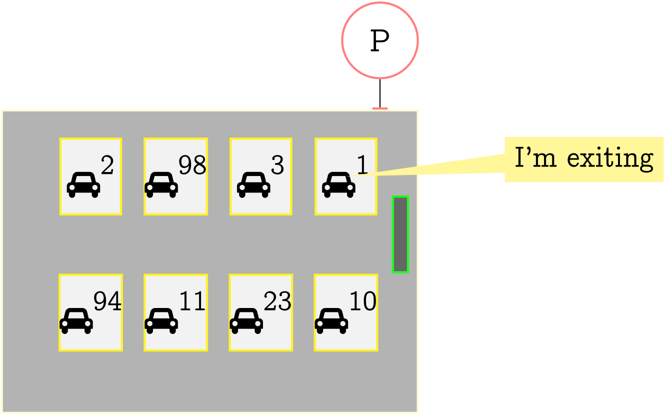

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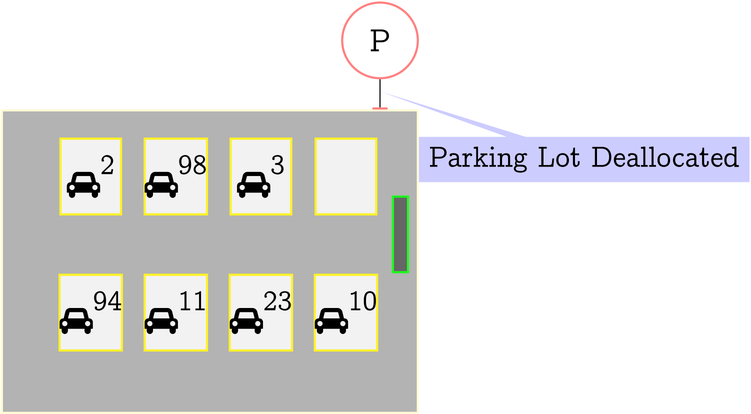

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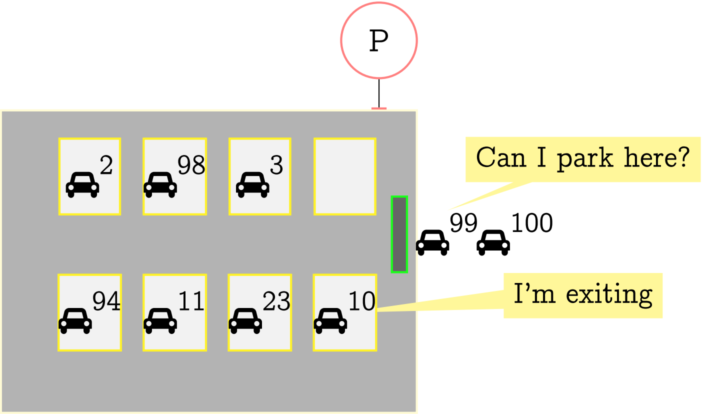
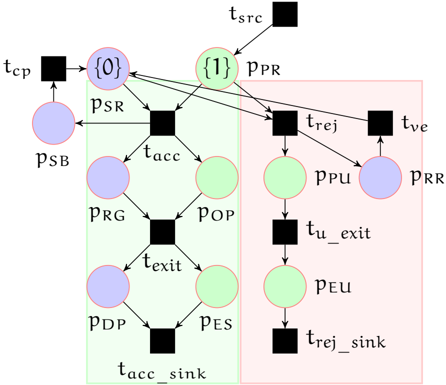


Modelling an Unbounded Client-Server (UCS)



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\end{aligned}
$$

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The proposed logic: The Monodic* Logic FOTL_1

The set of client formulae Δ:

$$\Delta ::= p(x) \mid \neg\alpha \mid \alpha \vee \beta \mid \alpha \wedge \beta \mid X_c\alpha \mid F_c\alpha \mid G_c\alpha \mid \alpha\ U_c\ \beta$$

where $\alpha, \beta \in \Delta$ and $p \in P_c$, the set of atomic client propositions.

The proposed logic: The Monodic* Logic $FOTL_1$

The set of client formulae Δ:

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where $\alpha, \beta \in \Delta$ and $p \in P_c$, the set of atomic client propositions.

The set of server formulae Ψ:

$$\begin{aligned}\Psi ::= & q \mid \neg\psi \mid (\exists x)\alpha \mid (\forall x)\alpha \mid \psi_1 \vee \psi_2 \mid \psi_1 \wedge \psi_2 \mid \\ & X_s\psi \mid F_s\psi \mid G_s\psi \mid \psi_1\ U_s\ \psi_2\end{aligned}$$

where $\psi, \psi_1, \psi_2 \in \Psi$ and $q \in P_s$, the set of atomic server propositions, $\alpha \in \Delta$.

The Monodic Logic FOTL_1 (contd)

$$\psi = (\exists x)(\exists y)\Big(\big(PR(x) \wedge PR(y)\big) \wedge F_c\big(ES(x) \wedge ES(y)\big)\Big)$$

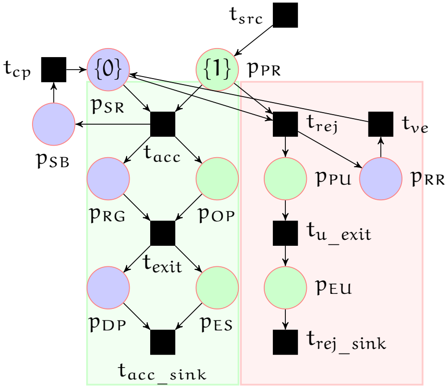
- Is $\psi \in \mathsf{FOTL}_1$?

The Monodic Logic FOTL_1 (contd)

$$\psi = (\exists x)(\exists y)\Big(\big(\mathsf{PR}(x) \wedge \mathsf{PR}(y)\big) \wedge \mathbf{F}_{\mathsf{c}}\big(\mathsf{ES}(x) \wedge \mathsf{ES}(y)\big)\Big)$$

- Is $\psi \in \mathsf{FOTL}_1$?
- Answer: No. **not a well formed formula** in FOTL_1

Recall: our model

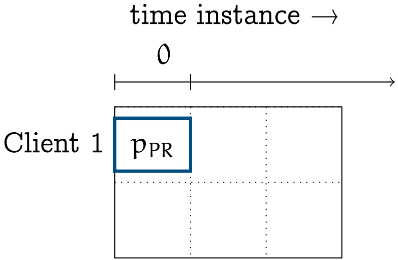


Atomic client propositions P_c and server propositions P_s:

$$P_c = \{\text{parking\_requested(PR)}, \text{occupy\_parking\_lot(OP)}, \text{parking\_unavailable(PU)}, \text{exit\_successfully(ES)}, \text{exited\_unsuccessfully(EU)}\}$$

$$P_s = \{\text{server\_ready(SR)}, \text{server\_busy(SB)}, \text{request\_granted(RG)}, \text{request\_rejected(RR)}, \text{deallocate\_parking\_lot(DP)}\}$$

Snapshots of the system + live windows



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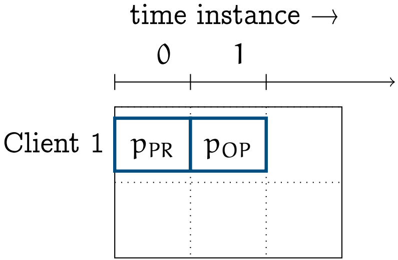

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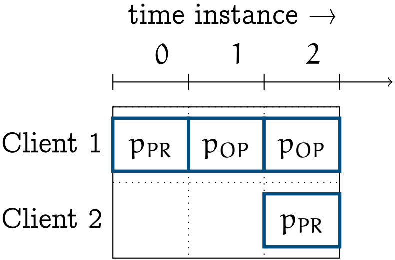
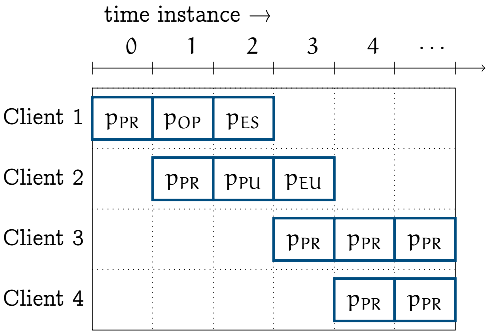


Snapshots of the system + live windows (contd)



- 4 distinguishable clients, with overlapping *live windows*, with the bound 5.
- client 1 is at p_{PR} at instance 0. i.e, $PR(x)$ is satisfied in client 1 at instance 0.
- For client 1, the *left boundary* is at instance 0 and its *right boundary* is at instance 2.

Formally speaking...

- Let CN be a countable set of client names.

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- Let $\mathfrak{Z} : CN \times \mathbb{N}_0 \rightarrow Q$ be a partial mapping describing the local state of each client $a \in CN$ at an instance $i \in \mathbb{N}_0$.

\- For instance, $\mathfrak{Z}(a, i) \in q$, means that the local state of each client a at instance i is state q, where $q \in Q$.

Formally speaking...

A model is a triple $M = (\nu, V, \xi)$ where

- ν gives the local behaviour of the server as follows:
  $\nu = \nu_0\nu_1\nu_2\ldots$, where for all $0 \leq i$, $\nu_i \subseteq P_s$

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- V gives the set of live agents (clients) at each instance.
  $V = V_0V_1V_2\ldots$, where for all $0 \leq i$, V_i is a finite subset of CN, gives the set of live agents at the ith instance.
  For every $0 \leq i$, V_i and V_{i+1} satisfy the following properties:
  1. if $V_i \subseteq V_{i+1}$ then for every $a \in V_{i+1} - V_i$ such that $\mathfrak{Z}(a, i+1) \in I$.
  2. if $V_{i+1} \subseteq V_i$ then for every $a \in V_i - V_{i+1}$ such that $\mathfrak{Z}(a, i) \in F$.

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- $\xi = \xi_0\xi_1\xi_2\cdots$, where for all $0 \leq i$, $\xi_i : V_i \to 2^{P_c}$ gives the properties satisfied by each live agent at ith instance. Also, $L(\mathfrak{Z}(a, i)) = \xi_i(a)$.

Semantics

$M, i \models (\exists x)\alpha$ iff $\exists a \in CN$, $a \in V_i$ and $M, [x \mapsto a], i \models_x \alpha$.

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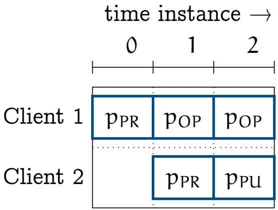

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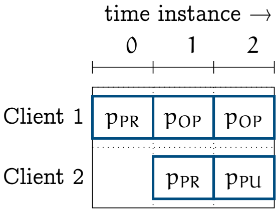

Given $\alpha = OP(x) \vee PR(x)$ and $CN = \{1, 2\}$.

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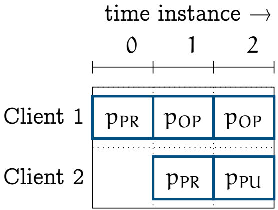
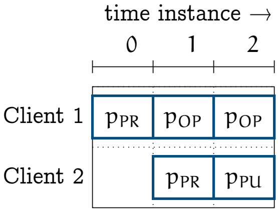


Given $\alpha = OP(x) \vee PR(x)$ and $CN = \{1, 2\}$.

- $M, 0 \models^k (\exists x)(OP(x) \vee PR(x))$.
- $M, 1 \models^k (\exists x)(OP(x) \vee PR(x))$.
- $M, 2 \models^k (\exists x)(OP(x) \vee PR(x))$.

Bounded Semantics

$$M, i \models^k (\forall x)\alpha \text{ iff } \forall a \in CN, \text{ if } a \in V_i \text{ then } M, [x \mapsto a], i \models^k_x \alpha$$



Given $\alpha = ES(x)$ and $CN = \{1, 2\}$.

- $M, 0 \not\models^k (\exists x)(ES(x))$.
- $M, 1 \not\models^k (\exists x)(ES(x))$.
- $M, 2 \not\models^k (\exists x)(ES(x))$.

Bounded Semantics

$M, i \models^k F_s\psi$ iff $\exists j : i \leq j \leq k$, $M, j \models^k \psi$.
Given $\psi = (\exists x)ES(x)$.

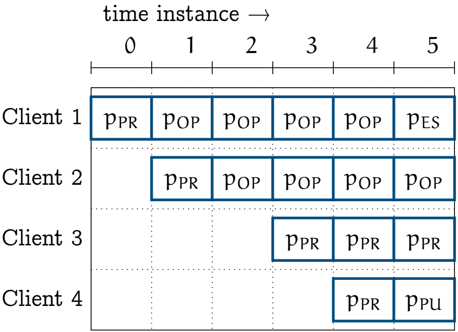


Figure: Snapshot with 4 clients

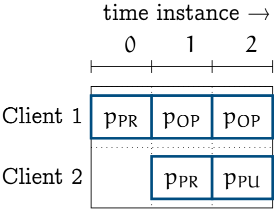


Figure: Snapshot with 2 clients

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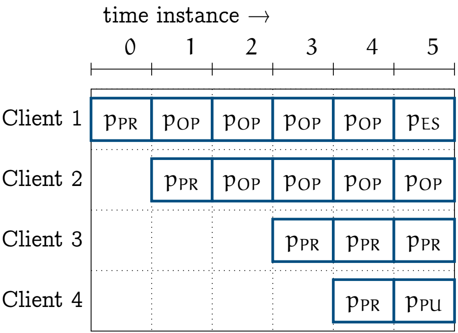

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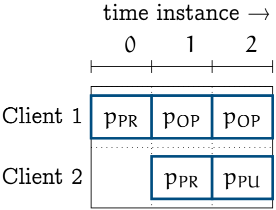

Figure: Snapshot with 2 clients

As shown above $M, 5 \models^k F_s(\exists x)ES(x)$.

Implementation

Verifying $G_s \exists x(F_c(p_0(x)))$

```
 evname: cToken_4_1
 evname: cToken_4_2
 evname: cToken_4_3
 evname: cToken_4_4
 evname: cToken_4_5
 evname: sToken_4_1
LOG: done with initializing empty token variables
LOG: sysObj.MAX_TOKENS_LIMIT before initial marking: 1
LOG: constructed initial Marking
LOG: constructed the complete Transition Function
LOG: nochange constraint added
LOG: UNSAT
LOG: UNSAT
LOG: UNSAT
LOG: UNSAT
LOG: --------------------------run 5 ----------------------------------
 evname: cToken_5_0
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```

Contributions

- Part 1: Verification of UCS using Petri Nets and $\mathrm{LTL}_{\mathrm{LIA}}$
- Part 2: Verification of UCS with distinguishable clients using ν-nets and properties specified in FOTL_1

Challenge 3: Modelling an UCS with internal structure

What if clients had an internal structure?

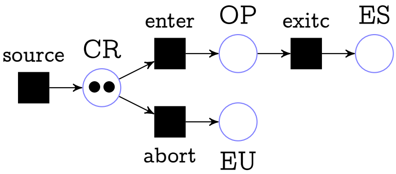


Figure: Petri net N_3 depicting client behaviour

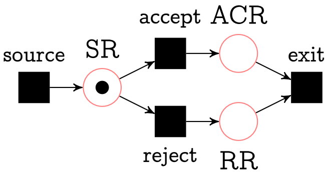


Figure: Petri net N_3 depicting server behaviour

Modelling UCS via Elementary Object Systems

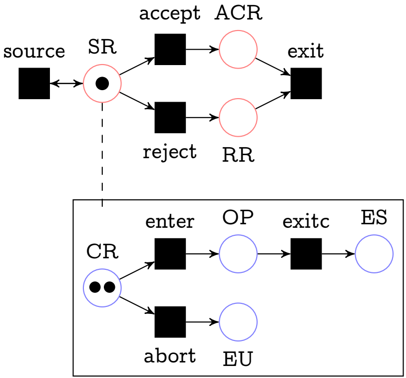


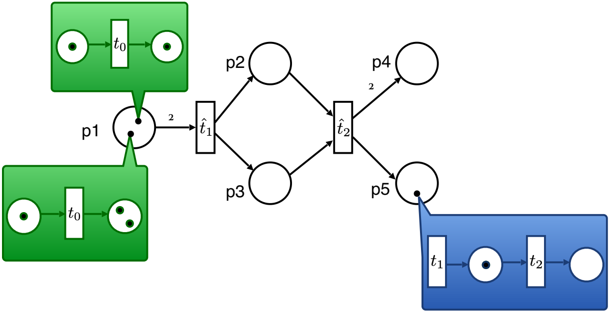
Figure: EOS modeling the single server (unbounded) multiple client system

Question: What properties of the system are we interested in?

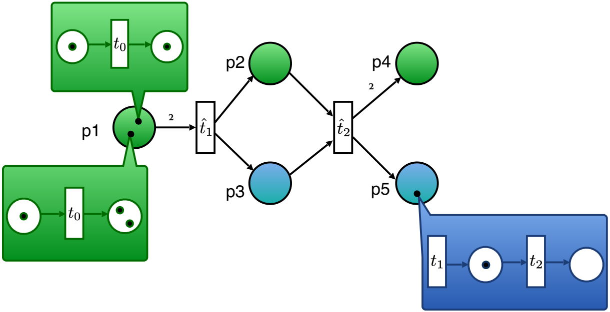
- **Reachability:** Is $\mu_1 = \langle \mathtt{SR}, \{\!\{\mathtt{client1CR}, \mathtt{client2CR}\}\!\} \rangle$ a reachable marking?
- **Coverability:** Given $\mu_1 = \langle \mathtt{SR}, \{\!\{\mathtt{client1CR}, \mathtt{client2OP}\}\!\} \rangle$ and $\mu_2 = \langle \mathtt{SR}, \{\!\{\mathtt{client1OP}, \mathtt{client2CR}, \mathtt{client3OP}\}\!\} \rangle$.

  Then, is $\mu_1 \preceq \mu_2$?

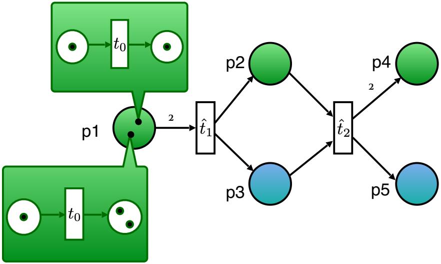
EOS – nested markings



EOS – typed places



EOS – object autonomous events



EVENTS

$$e_1 = \left(id_{p_1}, t_0\right)$$

$$e_2 = \left(\hat{t}_1, \varnothing\right)$$

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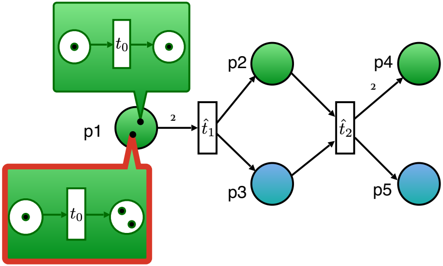

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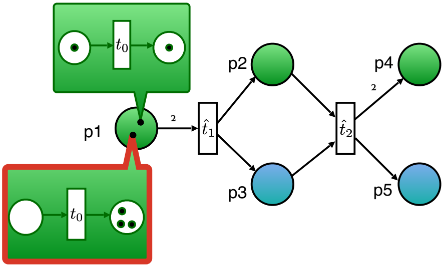

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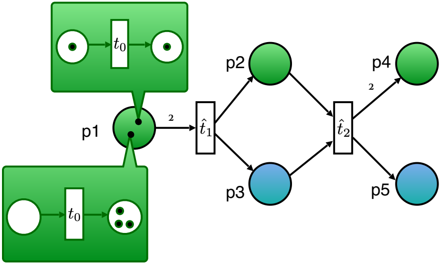

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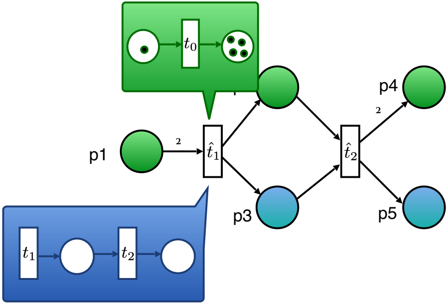

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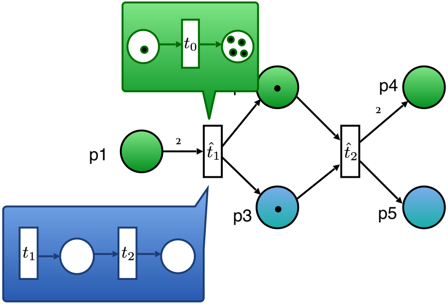

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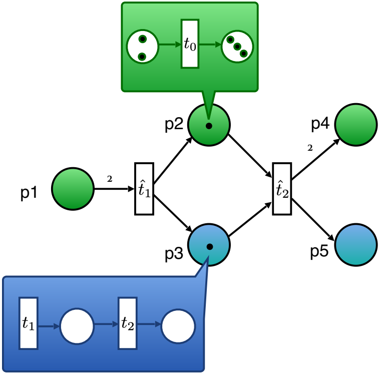

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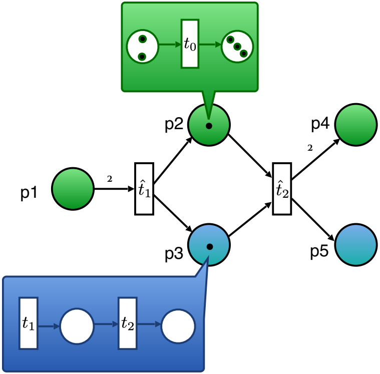
EOS – synchronization events



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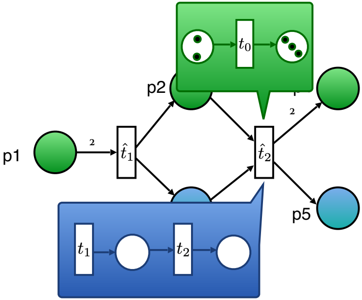

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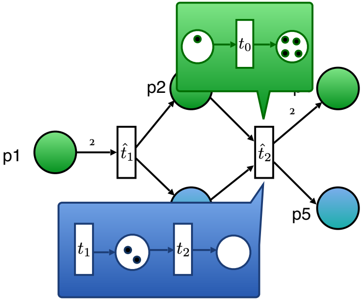

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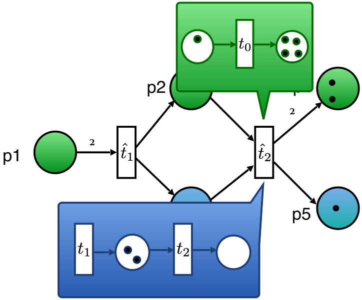

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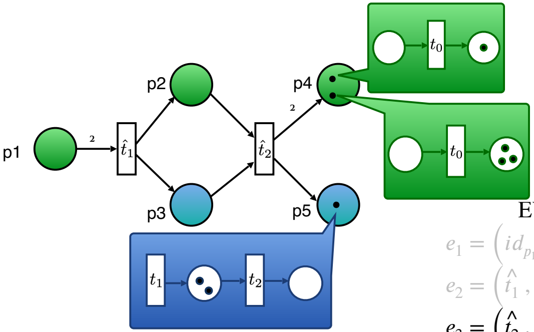

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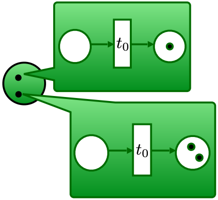
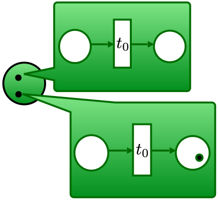
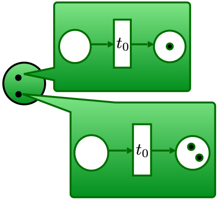
Object lossiness



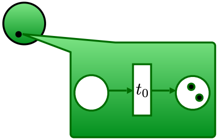
$$\geq_o$$



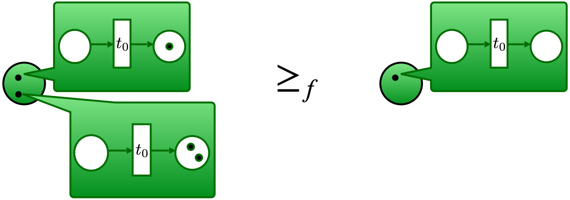
System lossiness



$\geq_s$



Full lossiness



$(\preccurlyeq, \ell)$-lossy runs

Perfect runs: only standard steps

$M_1 \to M_2 \to M_3 \to \dots$

$(\preccurlyeq, \ell)$-runs: at most $\ell \leq |\mathbb{N}|$ steps of type $\preccurlyeq$

$M_1 \to M_2 \succcurlyeq M'_3 \to M'_4 \succcurlyeq M'_5 \succcurlyeq M'_6 \to M_7 \to \dots$

$(\preccurlyeq, \ell)$-lossy problems

Is the system **robust** up to ℓ occurrences of $\preccurlyeq$?

$(\preccurlyeq, \ell)$-lossy problems

Is the system **robust** up to ℓ occurrences of $\preccurlyeq$?

$(\preccurlyeq, \ell)$-*deadlock freeness*

Input

An EOS E and an initial marking M.

Output

Is there a $(\preccurlyeq, \ell)$-run from M to a marking where no event is enabled?

Decidability Results (PNSE'24)

	Problems	$\preccurlyeq_f$	$\preccurlyeq_o$	$\preccurlyeq_s$
Conservative EOS	0-reach	**Undec** [Buß14]	**Undec** [Buß14]	**Undec** [Buß14]
	cover	**Dec** [Buß14]	**Undec** (2CM)	**Undec** (2CM)
	ℓ-reach/cover	**Dec** (Comp.)	**Undec** (Comp.)	**Undec** (Comp.)
	ω-reach/cover	**Dec** (Comp.)	**Undec** (Comp.)	**Undec** (Comp.)
EOS	0-reach	**Undec** [Buß14]	**Undec** [Buß14]	**Undec** [Buß14]
	cover	**Undec** [Buß14]	**Undec** (cEOS)	**Undec** (cEOS)
	ℓ-reach/cover	**Undec** (Gadget)	**Undec** (cEOS)	**Undec** (Comp.)
	ω-reach/cover	**Dec** (WSTS)	**Undec** (cEOS)	**Undec** (Comp.)

[Buß14] Köhler-Bußmeier, Michael. 'A Survey of Decidability Results for Elementary Object Systems'. 1 Jan. 2014 : 99 – 123.

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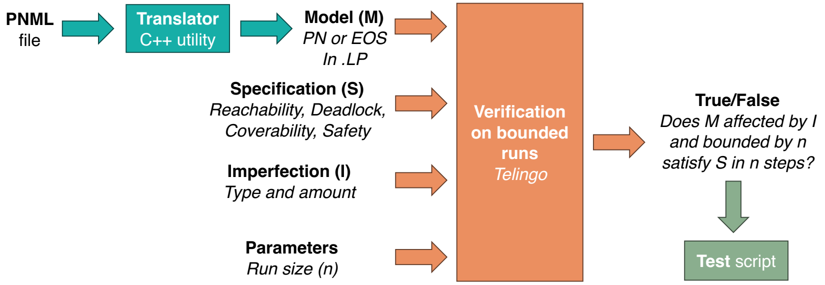
Motivation

Current PN tools lack of

- EOS support
- Imperfection support
- Robustness analysis
- Partial verification

Prototype (CILC'24)



Why bounded verification?

Problems on general EOS	$\preccurlyeq_f$	$\preccurlyeq_o$	$\preccurlyeq_s$
0-reach	U	U	U
0-cover	U	U	U
ℓ-reach/cover	U	U	U
ω-reach/cover	D	U	U

F. Di Cosmo, S. Mal, T. Prince, *Deciding Reachability and Coverability in Lossy EOS*, PNSE'24

Why Telingo?

Telingo is **declarative** and supports **temporal constraints**

- E.g., `:- &tel(>? (lossy >(>? lossy)` allows at most one lossy step
- The meaning of lossy is declared orthogonally to EOS specification

Telingo **returns finite runs**

- Perfectly matches bounded verification

Encoding of PNs and EOSs is **elegant in ASP**

- E.g., when compared to SMT – R. Phawade, T. Prince, S. Sheerazuddin et al., *Bounded Model Checking for Unbounded Client Server Systems,* Arxiv (2022)

Correctness and performances

Correct answers

- Checked on MCC benchmarks

Slow on PNs

- Compared with Tapaal

Prohibitively slow on EOSs

- Nesting exacerbates grounding

problem	lossiness	−imax =5 (s)	−imax =10 (s)	−imax =20 (s)	TAPAAL (s)
deadlock	none	UNSAT in 0.052	SAT in 0.622	SAT in 82.754	SAT in $5e-6$
deadlock	any	SAT in 0.009	SAT in 0.010	SAT in 0.009	NA
1-safeness	none	UNSAT in 0.010	UNSAT in 0.014	UNSAT in 0.027	UNSAT in 0
1-safeness	any	UNSAT in 0.013	UNSAT in 0.018	UNSAT in 0.032	NA

Table 1
Comparative Results with TAPAAL for the Eratosthenes-PT-010 PN from the MCC benchmarks [12].

Contributions

- Part 1: Verification of Unbounded Client-Server (UCS) using Petri Nets and Linear Temporal Logic with integer arithmetic
- Part 2: Verification of UCS with distinguishable clients using ν-nets and properties specified in monodic First Order Logic
- Part 3: Verification of UCS where the clients have internal structure using Elementary Object Systems and properties specified in Linear Temporal Logic

List of publications:

1. Ramchandra Phawade, Tephilla Prince, S. Sheerazuddin: Bounded Model Checking for Unbounded Client Server Systems. Arxiv (2026), submitted to TopNOC
2. Ramchandra Phawade, Tephilla Prince, S. Sheerazuddin: Verification of Unbounded Client-Server Systems with Distinguishable Clients Arxiv (2026), submitted to TopNOC
3. Tephilla Prince: On Verifying Unbounded Client-Server Systems. Springer EUMAS 2023: 465-471
4. Francesco Di Cosmo, Tephilla Prince: Bounded Verification of Petri Nets and EOSs using Telingo: An Experience Report. CILC 2024

List of publications:

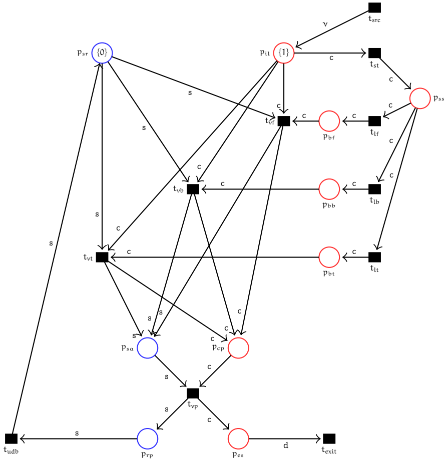
5 Francesco Di Cosmo, Soumodev Mal, Tephilla Prince: Deciding Reachability and Coverability in Lossy EOS. PNSE@Petri Nets 2024: 74-95

6 Francesco Di Cosmo, Soumodev Mal, Tephilla Prince: Decidability and Complexity of Reachability and Coverability in Lossy EOS (Yet to be submitted to Fundamenta Informatica)

List of open source tools built

1. DCModelChecker verifies PN over counting LTL properties
   https://doi.org/10.6084/m9.figshare.19875226
2. DCModelChecker1.0 verifies PN over $\mathrm{LTL}_{\mathrm{LIA}}$ with interleaved semantics
   https://doi.org/10.5281/zenodo.7084996
3. DCModelChecker2.0 verifies PN over $\mathrm{LTL}_{\mathrm{LIA}}$ allowing concurrent semantics
   https://doi.org/10.5281/zenodo.7198485
4. UCSChekcer verifies ν-nets over FOTL_1 properties
   https://doi.org/10.5281/zenodo.15847737
5. NWN Telingo Analyser: verifies safety, deadlock, reachability of lossy and perfect PN and EOSs
   https://doi.org/10.5281/zenodo.11401876

Appendix



t_{src}
v
p_{sr}
{0}
p_{il}
{1}
c
t_{st}
p_{ss}
s
t_{vf}
p_{bf}
t_{lf}
t_{vb}
p_{bb}
t_{lb}
t_{vt}
p_{bt}
t_{lt}
p_{sa}
p_{cp}
t_{vp}
t_{udb}
p_{rp}
p_{es}
d
t_{exit}

Bounded Semantics

$M, [x \mapsto a], i \models^k_x F_c \alpha$ iff
$\exists j : i \leq j \leq k,\ a \in V_j$ and $M, [x \mapsto a], j \models^k_x \alpha$.

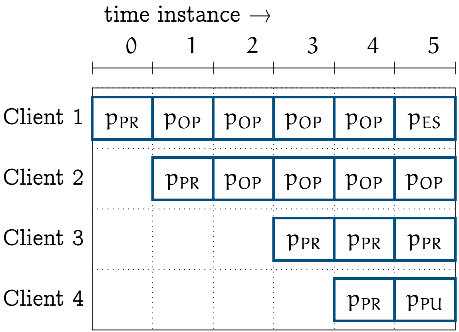


Figure: Snapshot with 4 clients

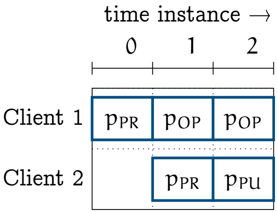


Figure: Snapshot with 2 clients

Given $\alpha = PR(x)$. The above formula is satisfiable for all clients in both figures.